DSA210 Project Proposal

This project investigates how team composition, specifically the distribution of champion roles, influences match outcomes in League of Legends. The main research question is whether the role structure of a team significantly affects the probability of winning a ranked match. The data will be obtained from a publicly available ranked-match dataset on Kaggle, which contains tens of thousands of match records (50,000+ samples). Each sample corresponds to a single ranked match and includes the champion IDs selected by both teams together with the final match outcome.

The data collection process will involve downloading the Kaggle CSV dataset and preprocessing it in Python. Since the raw dataset only provides champion IDs, additional champion metadata will be collected from Riot Games' Data Dragon API, which provides champion names and role information. These IDs will be matched with their corresponding roles through an automated mapping process. For champions associated with multiple roles, the first listed role will be assigned as the primary role to ensure consistency and avoid duplicate counting. After preprocessing, the data will be transformed into team-level samples, where each team is represented by the count of roles such as Marksman, Tank, Mage, Assassin, Support, and Fighter.

The final dataset will therefore consist of numerical features representing role distribution, role diversity, and specific role combinations, with the target variable being match win/loss outcome. The analysis will begin with exploratory data analysis to identify common winning role patterns, followed by statistical testing and basic machine learning models to evaluate whether team composition alone has predictive power. By focusing only on pre-game draft information, the project aims to provide interpretable insights into how champion-role balance contributes to competitive success.

Data Sources

- https://www.kaggle.com/datasets/datasnaek/league-of-legends
- https://ddragon.leagueoflegends.com/cdn/13.1.1/data/en_US/champion.json

Ayşe Çamlı - 34456